INORGANIC CHEMISTRY CUMULATIVE EXAM

December 2, 1995

d Orbitals

1. Give separate sketches for each of the five d orbital electron density clouds on labeled xyz coordinate axes. Name each.

2. Give the splitting pattern (with orbital name labels for each d level) for the d orbital energies of Pt in $PtCl_4^{-2}$.

3. Explain how d orbitals can be used in accounting for the molecular geometry (and relative bond lengths) in SF_4.

4. How are d orbitals used in explaining the W-C bonding in tungsten hexacarbonyl?

5. What are the two different d-orbital explanations for P-N bonding in $(NPCl_2)_3$?

6. How are d orbitals used in the description of the metal-metal quadruple bond?

7. How might the ligating ability of triphenyl phosphine involve the d orbitals of P?

8. La, Lu, and Gd all have an electron configuration which includes the notation $5d^1$. Explain why each does.

9. How are d orbitals used to explain part of the differences in chemical reactivity of disilane and ethane?